Beneath the Surface: Catching Cartels in the Act in Brazilian Procurement*

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Abstract

Brazil's antitrust authority secured administrative convictions against 35 firms in seven procurement cartels operating in the state of São Paulo between 2009 and 2019. We match these firms and the product markets delineated in the case files to 3.8 million bids on the state's electronic procurement platform and to the universe of matched employer–employee records, assembling the first conviction-anchored benchmark that integrates auction behavior, labor microdata, and antitrust rulings.

On a unified benchmark of roughly 2,300 close-margin pregão auctions (absolute bid margin $|MV| \leq 0.10$) in which at least one convicted firm participated during the active period of its cartel, canonical variance-based screens fire in the direction predicted by the bid-rotation model—bid dispersion and the coefficient of variation are weakly compressed (Cohen's $d \approx -0.07$ to -0.10) and the number of active bidders is modestly lower ($d \approx -0.24$)—yet the corresponding areas under the ROC curve (0.52–0.56) remain far too low for investigative triage, indicating near-total distributional overlap between cartel and clean auctions. A close-margin regression discontinuity in the spirit of Kawai et al. (2023) likewise yields a null. By contrast, network screens constructed from the same benchmark achieve markedly higher discrimination: inter-firm worker flows from the matched employer–employee data yield AUCs of 0.75 (shared workers) and 0.74 (Jaccard similarity); a co-bidding density screen yields 0.70. The worker-flow signal survives conditioning on sector and on co-bidding intensity. A specialization placebo restricted to non-cartel firm pairs bidding on the same product code in the same year

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absorbs approximately half of the worker-flow signal, leaving a cartel-specific residual—consistent with cartels both inheriting and reinforcing the personnel flows natural to thin specialized-labor markets.

These findings support a view in which collusion leaves institution-specific signatures: in markets with thin specialized-labor pools, the cheapest coordination technology operates through personnel flows rather than bid-pattern manipulation. For competition authorities with access to employer–employee microdata, worker-flow screens offer a deployable complement to canonical bid-pattern tests.

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1. Introduction

Between 2009 and 2019, Brazil's federal antitrust authority—the *Conselho Administrativo de Defesa Econômica* (CADE)—secured administrative convictions against 35 firms across seven distinct procurement cartels operating in the state of São Paulo, with case files delineating the specific product markets and time windows over which each ring was active. We match these convicted firms and tender windows to the universe of 3.8 million bids placed on São Paulo's electronic procurement platform, the *Bolsa Eletrônica de Compras* (BEC), and to the universe of formal-sector employment relationships recorded in the Relação Anual de Informações Sociais (RAIS). On the resulting benchmark of roughly 2,300 close-margin auctions in which at least one convicted firm bid during the active period of its cartel, the canonical variance-based screens of Porter and Zona (1993) and Bajari and Ye (2003) behave in the direction predicted by the bid-rotation model: bid dispersion, the coefficient of variation of bids, and the number of active bidders are all weakly compressed at cartel-tainted auctions (Cohen's $d \in [-0.07, -0.24]$). Yet the areas under the receiver-operating-characteristic curve (AUCs) for these screens range from 0.52 to 0.56—above chance but far too low for investigative triage. A close-margin regression discontinuity in the spirit of Kawai et al. (2023) likewise produces a null. By contrast, network screens constructed from inter-firm worker flows in the matched employer–employee data recover AUCs of 0.75 (shared workers) and 0.74 (Jaccard similarity), and a co-bidding density screen yields 0.70, on the same benchmark. A specialization placebo restricted to same-market non-cartel firm pairs absorbs approximately half of this signal, leaving a cartel-specific residual that indicates cartels both inherit and reinforce the personnel flows natural to thin specialized-labor markets. This paper documents these facts, interrogates their mechanism, and draws out the implications for the design of cartel-detection screens in such markets.

A growing literature uses statistical regularities in bidding behavior to infer collusion that is never documented directly in the administrative record. Classical screens exploit the compression induced by bid-rotation agreements: colluding firms submit prearranged prices that cluster tightly around the designated winner, yielding low dispersion, few active bidders, and predictable rotation patterns (Porter and Zona, 1993; Pesendorfer, 2000; Bajari and Ye, 2003; Conley and Decarolis, 2016; Imhof et al., 2018). A complementary and influential approach, recently advanced by Kawai et al. (2023), focuses on the winning–losing threshold: under competition, the observable characteristics of narrowly

winning and narrowly losing firms should be continuous at a bid margin of zero, and any statistically sharp discontinuity at the cutoff identifies an anti-competitive wedge. Both approaches have been validated primarily against cases in high-income economies with deep formal labor markets, dense bidder pools, and strong institutional capacity for antitrust enforcement; their extension to thinner institutional environments rests on a small but growing empirical base. Wachs and Kertész (2019) apply a co-bidding network approach to roughly 150,000 Georgian procurement notices and show that group-level cohesion and exclusivity flag candidate cartels in the absence of criminal ground truth. Signor et al. (2020) exploit the dataset released by the Brazilian Federal Police's *Operação Lava Jato* investigation to build a probabilistic detector of collusive tenders in the Petrobras construction-cartel case, reporting accuracy between 81 and 96 percent. Huber and Imhof (2019) combine classical statistical screens with supervised machine-learning classifiers and correctly predict 84 percent of Swiss tenders out of sample, demonstrating the gains from combining bid-distribution screens with flexible functional forms. What has been missing is a direct test against an independent criminal ground truth in a setting with *both* structurally different institutional primitives *and* access to administrative employer–employee microdata. The São Paulo–CADE–RAIS linkage we build offers exactly that test, and is, to our knowledge, the first cartel-detection benchmark to integrate auction-level behavioral data, employer–employee matched labor records, and formal antitrust convictions. The policy stakes are first-order: middle-income competition authorities increasingly rely on off-the-shelf screens developed and validated abroad to triage investigations (Organisation for Economic Co-operation and Development, 2015; Bosio et al., 2022), and if those screens systematically mislabel cartels under local conditions, existing enforcement pipelines may be misfiring in precisely the settings where deterrence capacity is weakest.

Our primary benchmark is a *cartel-active* sample of roughly 2,300 close-margin pregão auctions (with absolute bid margin $|\mathrm{MV}| \leq 0.10$) in which at least one convicted firm participated during the active period of its cartel. This is the sample on which classical, regression-discontinuity, and network-based screens can all be estimated with enough statistical power to discriminate cartel auctions from a clean control. We supplement it with a *strict pair-matched* sample of 14 auctions in which both the winning and the runner-up firms belong to the same convicted cartel during its prosecuted conduct window. The strict sample is informative about cartel mechanics in head-to-head competition in principle, but is severely power-limited. For transparency we also report results

on an *anytime-listed* sensitivity sample of 107 auctions in which both the winner and the runner-up appear in a CADE conviction for the same cartel process but the auction year may fall outside the prosecuted conduct window—of which 93 are out-of-window auctions and 100 of the 107 come from a single cartel (the medicamentos hub-and-spoke case, CADE process 08012.002222/2011-09). Because whether out-of-window bidding by firms that were cartelists in another year should be treated as cartel behavior depends on assumptions this paper does not attempt to resolve, the anytime sample is reported only as a sensitivity and is not the basis for any headline claim. We refer to the three samples collectively as *conviction-anchored detection benchmarks*.

Our first set of results documents the limited discriminative power of canonical screens in this institutional setting. On the unified cartel-active benchmark of roughly 2,300 close-margin pregão auctions ($|MV| \leq 0.10$), classical variance screens fire in the direction predicted by the bid-rotation model: bid dispersion and the coefficient of variation are weakly compressed at cartel-tainted auctions relative to clean controls (Cohen's $d \approx -0.07$ to -0.10), and the number of active bidders is modestly lower ($d \approx -0.24$), with Welch t-statistics that are statistically significant given the large control sample. Yet the effect sizes are small and the distributional overlap between cartel and clean auctions is near-total: the areas under the receiver-operating-characteristic curves (AUCs) for these screens range from 0.52 (coefficient of variation) to 0.56 (bidder count)—above chance but far too low for investigative triage. A Wilcoxon rank-sum test confirms the weak rank-order separation (rank-sum $p < 10^{-4}$ for each, aligning with the AUC direction). Classical screens are therefore directionally consistent with theory but operationally inadequate as cartel detectors in this setting: a triage tool built on these statistics would generate an unacceptable rate of false positives and false negatives relative to the conviction-anchored ground truth. The same qualitative pattern holds on the strict pair-matched sample of 14 auctions—canonical direction, low discriminative power, and mostly non-significant individual t-tests owing to small-N constraints. A direct replication of the Kawai close-margin RD on the cartel-active subsample yields no statistically detectable discontinuity in incumbency or last-bid behavior at the cutoff, and a density test in the spirit of Imhof et al. (2018) detects no manipulation of the running variable. The anytime-listed sensitivity sample of 107 auctions does produce positive Δ estimates on variance screens with $p < 0.02$, but we interpret this as a mechanical consequence of the 93 out-of-window auctions in that sample—auctions in which firms that had been cartelists in another year are bidding against each other under competi-

tive conditions—rather than as evidence that cartel auctions exhibit elevated dispersion. In short, while canonical variance-based screens exhibit weak directional signal, neither they nor the close-margin RD of Kawai et al. (2023) achieve the discriminative power required to operationally detect the cartels that CADE has formally convicted.

Our second set of results shows that network-based screens constructed from administrative labor data recover strong and statistically robust cartel detection on the cartel-active benchmark. For each auction, we construct two network screens from the five-year window of RAIS records preceding the tender: the number of workers who have moved between the winner and the runner-up firms, and the Jaccard similarity of the two firms' historical workforces. On the cartel-active sample of roughly 2,300 close-margin auctions, these screens yield AUCs of 0.750 and 0.739 respectively—outperforming the best classical screen by roughly 20 percentage points and representing, to our knowledge, the first robust AUCs reported for procurement cartel detection against a formal criminal ground truth. A third screen, built from the co-bidding network of shared tender participations, yields an AUC of 0.695, providing independent corroboration through a distinct information channel. Critically, the worker-flow signal is neither an artifact of sectoral composition nor a mechanical re-expression of co-bidding intensity. Restricting attention to firm pairs that share the same 2-digit sectoral classification reduces the worker-flow AUC only marginally, from 0.739 to 0.730, and the residual of worker-flow after partialling out the log co-bidding density of each pair achieves an AUC of 0.749—essentially unchanged from the raw baseline, indicating that worker mobility carries detection information above and beyond what co-bidding patterns alone can reveal. A more demanding test separates cartel-specific coordination from passive same-market labor circulation: we construct a *specialization placebo* of non-cartel firm pairs bidding on the same BEC item code in the same year as each cartel-active auction, and compare worker-flow distributions across cartel, placebo, and global clean samples. Cartel pairs share at least one worker in 57.9% of cases, compared with 31.1% for same-market placebo pairs and 9.9% for pairs in other markets. In a decomposition of the worker-flow AUC gain relative to the global clean baseline, roughly half is absorbed by the same-market specialization component and roughly half remains as a cartel-specific residual. Both components are consistent with our institutional-thinness mechanism: cartels inherit the labor-circulation patterns natural to thin specialized-labor markets and additionally reinforce those flows to sustain coordination. On the strict pair-matched sample ($N = 10$ after close-margin filtering), worker-flow AUCs decline to near chance; we at-

tribute this to power constraints at small sample sizes rather than to any breakdown in the underlying signal, and we emphasize that the cartel-active benchmark is where the discriminative power of any screen—classical or network-based—can be meaningfully evaluated. The conviction-anchored benchmark does not merely declare cartels detectable in São Paulo; it pinpoints the channel through which the detection signal propagates—worker mobility, not bid dispersion.

We interpret these findings through the lens of institutional thinness. In settings where the universe of firms capable of bidding on a given product is small and the formal-sector labor market for the specialized workers that serve those bids is shallow, coordination among colluding firms generates naturally detectable traces in worker flows: personnel rotations, key-employee transfers, and shared specialized labor become vehicles for, and therefore signatures of, the collusive communication that substitutes for the statistical-pattern coordination captured by canonical screens in thicker markets. This mechanism—which we develop formally in a simple model in Section 4—rationalizes both the low discriminative power of bid-dispersion screens and the success of network-based screens in the Brazilian setting. Critically, it also predicts the two-component decomposition of the worker-flow signal documented by our specialization placebo: because thin markets generate spontaneous labor circulation among same-market firms whether or not they are colluding, a non-cartel placebo restricted to same-market firm pairs should already exhibit elevated worker-flow overlap relative to random pairs; and cartel pairs should exhibit still greater overlap on top of that baseline. Both predictions are borne out in the data. The mechanism further predicts that classical screen-based enforcement strategies will systematically under-detect cartels in precisely the institutional environments where their deterrence value would be highest.

This paper makes three contributions. *First*, methodologically, we deliver the first stress test of two complementary cartel-detection frameworks against an independent criminal ground truth, and we document a common failure mode: low discriminative power. On the unified cartel-active benchmark, canonical variance-based screens (Porter and Zona, 1993; Bajari and Ye, 2003) fire in the direction predicted by the bid-rotation model—compression of bid dispersion and bidder counts (Cohen's $d \in [-0.07, -0.24]$)—but yield AUCs of only 0.52–0.56, far below the threshold required for operational triage. The close-margin RD design of Kawai et al. (2023) similarly produces a null. Neither result repudiates the underlying designs, both of which remain internally valid; rather, both warn that the *operational discriminating power* of screens developed against car-

tels in thick-labor environments is not invariant when these tools are redeployed to settings in which cartel coordination is suspected to rely on alternative technologies such as labor mobility. *Second*, substantively, we introduce conviction-anchored detection benchmarks by linking CADE rulings, BEC bid-level records, and RAIS employer–employee data, and we show that worker-flow and co-bidding network screens constructed from these sources substantially outperform classical screens in detecting convicted cartels. Because such data are increasingly available to national competition authorities in middle-income economies, the screens we propose are directly deployable at negligible marginal cost. *Third*, conceptually, we advance the view that collusion leaves institution-specific signatures, and that the design of an effective detection screen should depend on which coordination technology is cheapest for cartel members to operate in the relevant environment—a framing that organizes and partially reconciles the growing patchwork of contradictory empirical findings in the detection literature (Conley and Decarolis, 2016; Imhof et al., 2018; Kawai et al., 2023).

The remainder of the paper is structured as follows. Section 3 describes the institutional setting of São Paulo's public procurement and the CADE cartel cases that anchor our ground truth. Section 2 situates our contribution in the procurement-collusion detection literature and in the adjacent literatures on worker mobility and law-and-economics enforcement. Section 5 describes the BEC, RAIS, and CADE data sources and the construction of the cartel-active and pair-matched benchmark samples. Section 4 presents our empirical strategy—the RD replication, the portfolio of classical screens, and the network-based screens—and sketches a simple theoretical framework linking screen predictions to institutional primitives. Section 6 reports the main results. Section 8 discusses policy implications and concludes.

2. Related Literature

This paper engages three distinct literatures: cartel detection via bid-pattern screens, network-based approaches to collusion identification, and the economics of inter-firm worker mobility.

2.1. Detection through bid-pattern screens

A foundational literature exploits the prediction that bid-rotation agreements compress the statistical properties of bids. Porter and Zona (1993) introduced variance-based

screens to the railroad freight cartel of the 1880s, showing that collusive phases exhibit lower bid dispersion than competitive phases when the cartel's internal records provide an exogenous regime switch. Pesendorfer (2000) documented bid rotation in Florida school-milk procurement, demonstrating that cartel members systematically coordinate winning turns. Bajari and Ye (2003) proposed exchangeability tests for bidder costs in sealed-bid auctions, applying them to highway paving contracts in the U.S. Midwest; rejections of exchangeability flag candidate collusive pairs. Conley and Decarolis (2016) extended these ideas to Italian procurement, using pair-level bid-correlation tests to detect bidder groups. Imhof et al. (2018) combined classical variance and bid-level screens with density tests in a Swiss procurement setting, demonstrating that multi-screen portfolios improve detection relative to any single indicator. Abrantes-Metz et al. (2006) provided practical guidance on implementing price-fixing screens, cautioning against single-indicator approaches. Harrington (2008) synthesized the design principles for detection mechanisms, emphasizing that the informativeness of any screen depends on the competitive counterfactual and the cartel's coordination technology—a point our findings make concrete.

More recently, Kawai et al. (2023) proposed a regression discontinuity design centered on the winning–losing margin in Japanese construction procurement: under competition, firm characteristics should be continuous at the bid margin of zero, and any sharp discontinuity identifies a non-competitive wedge. Their approach has high internal validity but, as our results show, may lack discriminative power in settings where collusion does not concentrate at the bid margin. Chassang and Ortner (2019) offered a complementary theoretical framework in which bid constraints endogenously shape the collusive technology, predicting that auction-format-specific screens may outperform generic variance tests—a prediction consistent with our finding that classical screens behave differently across pregão and convite formats.

A common thread in this literature is that almost all validation exercises rely on ground truth from high-income economies with deep bidder pools and strong enforcement capacity. Levenstein and Suslow (2006) documented the institutional determinants of cartel success and duration across 81 international cartels, noting that market structure, transparency, and enforcement intensity jointly shape collusive outcomes. What remains scarce is direct validation of off-the-shelf screens against criminal ground truth in middle-income settings with structurally different labor markets and thinner bidder pools—precisely the gap our conviction-anchored benchmark addresses.

2.2. Network-based and machine-learning approaches

A growing literature moves beyond bid distributions to exploit relational and network information. Wachs and Kertész (2019) applied co-bidding network analysis to roughly 150,000 Georgian procurement notices, showing that group-level cohesion and exclusivity metrics flag candidate cartels even in the absence of criminal ground truth. Signor et al. (2020) exploited the dataset released by Brazil's *Operação Lava Jato* investigation to build a probabilistic detector of collusive tenders in the Petrobras construction-cartel case, reporting accuracy between 81 and 96 percent. Huber and Imhof (2019) combined classical statistical screens with supervised machine-learning classifiers and correctly predicted 84 percent of Swiss tenders out of sample, demonstrating the gains from combining bid-distribution features with flexible functional forms. Clark et al. (2021) studied collusion and corruption jointly in Québec construction procurement, documenting how collusive networks co-evolve with political connections and how entry deterrence interacts with bid rigging.

These network approaches share a common insight: collusion is fundamentally a relational phenomenon, and screens that exploit the structure of firm–firm interactions can extract signal that univariate bid statistics miss. Our contribution builds on this insight but shifts the network channel from co-bidding patterns (observable within the procurement data itself) to worker mobility across firms (observable only in matched employer–employee administrative records). This shift is consequential because worker-flow networks are constructed from an entirely independent data source, reducing the risk of circularity between the screen and the behavioral patterns it is designed to detect.

2.3. Worker mobility as an inter-firm transmission channel

A separate literature in labor economics establishes that worker flows between firms serve as a vehicle for knowledge transmission, coordination, and productivity spillovers. Stoyanov and Zubanov (2012) showed that productivity gains propagate across Danish firms through the mobility of workers, with the magnitude of the spillover proportional to the productivity gap between sending and receiving firms. Serafinelli (2019) documented similar patterns in the Veneto region of Italy, demonstrating that workers moving from high-productivity firms raise wages and productivity at destination firms. Jarosch et al. (2021) developed an equilibrium model of learning from coworkers, showing that team composition and worker reallocation jointly determine firm-level human capital accumu-

lation. Together, these results establish that worker flows are not merely a labor-market phenomenon but an active channel of inter-firm information transmission.

We bring this insight to the collusion-detection setting. If worker mobility transmits knowledge and facilitates coordination between firms in general, it may also serve as a coordination technology for cartel members—and, critically, leave detectable traces in administrative records. In markets with thin specialized-labor pools, where the universe of workers with the requisite expertise is small, the signal-to-noise ratio of worker-flow screens should be highest: cartel-related personnel transfers stand out against a sparse baseline of spontaneous mobility. Our specialization placebo formalizes this intuition by decomposing the worker-flow detection signal into a same-market specialization component (present even among non-colluding firms in thin markets) and a cartel-specific residual (the incremental coordination signature). To our knowledge, ours is the first paper to deploy matched employer–employee data as a cartel-detection screen and to validate it against formal antitrust convictions.

3. Institutional background

Public procurement in the state of São Paulo is conducted through a decentralized system of purchasing agencies, each referred to as a Public Buyer Unit (PBU). Since 2007, the use of the state's electronic procurement platform, the *Bolsa Eletrônica de Compras* (BEC), has been mandatory for PBUs when acquiring common goods and services. The BEC platform centralizes tender procedures and records detailed information on each procurement process. It plays a crucial role in São Paulo's procurement operations: for instance, in 2018 alone, approximately US$1.7 billion worth of goods and services were procured via BEC. Over the period from 2009 onward, the BEC system handled more than R$7.4 billion (approximately 34% of all state government purchases) for the State Health Secretariat (SES/SP) and other agencies in São Paulo. Given the large scale and value of these tenders, ensuring fairness and preventing collusion in the procurement process is of paramount importance. A failed tender (one that ends with no supplier selected) not only delays public service delivery but also wastes significant administrative resources, as each tender process can cost the government between US$500 and US$5,200 in preparation and execution.

The BEC catalog contains a comprehensive listing of standardized items and services, classified by category. Nearly 55,000 purchase offers were issued through BEC during our

study period, covering about 6,350 distinct items ranging from medical supplies to office equipment. Each purchase offer (PO) is a digital solicitation document that identifies one or more items to be procured, along with the required quantities and specifications. Each item within a PO is auctioned separately; thus a unique procurement transaction can be defined by a combination of a PO number and an item code (POI). The BEC records include rich information for each POI, including the reference (reserve) price set by the buyer, the number of participating firms, all bids submitted (identifying winners and losers), the auction format used, and the identities of the buyer (PBU) and auctioneer.

São Paulo's PBUs predominantly use two types of competitive tendering procedures available on BEC: (i) a sealed-bid tender (*convite*) and (ii) a multi-round descending auction (*pregão*). In a sealed-bid tender, firms submit confidential bids by a specified deadline. After the submission period closes, all bids are opened and the contract is awarded to the qualified bidder with the lowest bid price. The sealed-bid format is typically used for purchases up to a value limit of R$176,000 (approximately US$35,000). In contrast, the descending auction (pregão) has no value limit and involves an open, interactive bidding phase. Initially, firms submit sealed proposals which are evaluated for qualification. Then, all qualified bids are announced (with the bidders' identities concealed) and an open auction phase commences. During this phase, which lasts for an initial 20 minutes, bidders can submit progressively lower bids, knowing the current lowest bid in real time. If a new bid is placed in the final minutes of this period (between 16 and 20 minutes), the auction is automatically extended by an additional 4 minutes. The auction concludes when no new bids arrive for 4 consecutive minutes. The winner is the bidder offering the lowest price at the end of this process, provided that price is below the government's preset reference price.

The key difference between the sealed-bid and descending auction formats lies in the opportunity for iterative bidding and post-auction negotiation. The pregão format, after determining the lowest bid through the auction, often allows a brief negotiation or confirmation step to potentially further reduce the price or clarify terms with the lowest bidder, as permitted by Brazilian procurement rules. In contrast, the convite (sealed bid) format is a single-shot process with no negotiation stage, which makes it simpler but also means the outcome is entirely determined at bid opening. From an oversight perspective, the open auction format provides more transparency during the bidding process (as all bids are observable in real time, albeit anonymously), whereas the sealed bid format concentrates all transparency at the moment of opening. Both mechanisms are subject

to anti-collusion regulations and audit processes, but their procedural differences can influence how collusion or favoritism might manifest. Our analysis will focus specifically on sealed bid auctions.

4. Identifying anti-competitive behaviors

Our empirical strategy follows the approach of Kawai et al. (2023) to detect subtle signs of collusion or favoritism in auction outcomes. The core idea is to focus on procurement auctions where the competition is extremely tight, meaning the winner's bid is nearly identical to the runner-up's bid. Under genuinely competitive conditions, two firms bidding almost the same price should be similar in their cost efficiency and should not systematically differ in their past bidding success. Thus, absent any collusion or external favoritism, one would expect no abrupt differences in firm characteristics when moving from just losing an auction to just winning it.

We implement this idea using a regression discontinuity design (RDD) centered around the margin of victory. Let $\Delta_{i,t}$ be the difference between firm i's bid and the lowest competing bid in auction t. This margin $\Delta_{i,t}$ serves as our running variable, with a cutoff at zero. Firms with $\Delta_{i,t} < 0$ win the auction (their bid is lower than the best alternative), whereas those with $\Delta_{i,t} > 0$ lose. Intuitively, $\Delta_{i,t} = 0$ represents a threshold between winning and losing. We focus on auctions where $|\Delta_{i,t}|$ is very small, meaning the winner barely underbid the closest rival. In such cases, any systematic jump in firm outcomes at $\Delta_{i,t} = 0$ can be interpreted as evidence of an outside influence on the auction result.

The two primary outcomes we examine are the firm's *incumbency status* and its bid timing, identifying whether it is the *last bidder* in the auction. We define an *incumbent* as a firm that won the last auction it participated in prior to the current auction (incumbency is an indicator variable equal to 1 if the firm was the previous winner and 0 otherwise). *Last bidder* identifies whether a firm was the last among the competitors to submit the bid. By examining these variables for firms just on either side of the winning threshold, we can detect irregular patterns: for instance, an unexpectedly high incumbency rate or unusually larger probability of being the last bidder for firms that barely win compared to those that barely lose.

Formally, we estimate the discontinuity at $\Delta = 0$ for these outcomes. Let $x_{i,t}$ denote a generic outcome (such as the incumbency indicator or backlog) for firm i before auction t.

The discontinuity is captured by the difference in the expected value of $x_{i,t}$ immediately to the left and right of the cutoff:

$$\beta \;=\; \lim_{\epsilon \to 0^+} \mathbb{E}[\, x_{i,t} \mid \Delta_{i,t} = \epsilon \,] \;-\; \lim_{\epsilon \to 0^-} \mathbb{E}[\, x_{i,t} \mid \Delta_{i,t} = \epsilon \,] \;,$$

where ϵ is a small positive increment in the bid margin. In the absence of any strategic behavior or external interference, we would expect $\beta \approx 0$ for characteristics like incumbency, since firms just winning or just losing should be comparable. Conversely, a significant $\beta \neq 0$ indicates a discontinuous jump in firm characteristics at the threshold, consistent with non-competitive forces affecting the outcome.

We hypothesize that a positive advantage conferred to narrowly winning firms would manifest as $\beta > 0$ in our setup (because the outcome for winners exceeds that for losers). Detecting a substantial gap in these measures at $\Delta = 0$ will allow us to distinguish between a competitive market and one influenced by collusion or favoritism.

For example, if there is political favoritism benefiting certain firms, those firms would appear as incumbents much more often when they win narrowly, whereas otherwise identical firms that narrowly lose would not show such a history. Similarly, a collusive bid-rotation scheme might lead to a more equal distribution of recent wins among participants, in which case we would *not* expect a sharp drop in backlog for losing firms at the threshold.

5. Data

5.1. Sources

We utilize a rich administrative dataset of public procurement auctions conducted via the BEC platform in São Paulo state. The dataset covers the period from January 2009 to December 2019 and includes detailed bid-level information for auctions of common goods and services. During this period, 1,344 PBUs (encompassing state-level executive agencies, the legislative and judicial branches, as well as some municipal governments and even a few private organizations authorized to use the system) carried out procurement through BEC. In total, the data comprise 3,866,407 individual bidding transactions submitted by 19,007 distinct firms, corresponding to 169,607 unique procurement items. Each record in the data identifies the purchase offer (PO) and item, the participating firm, the bid amount, and the auction outcome (win or loss), among other details.

The BEC system's hierarchical catalog organizes items into groups and classes. For example, medical supplies are grouped under a specific category (e.g., group 65 for medical and hospital items), with finer classifications down to individual item codes.[1] This structured coding is useful for analyzing procurement patterns by item type or sector. Our analysis leverages the full breadth of goods and services in the dataset, while controlling for any category-specific effects as needed.

For the purposes of our study, we focus on competitive tenders that resulted in a contract award (i.e., at least one winner and one or more losing bids). We exclude tenders that were canceled or received only a single bid, since those do not provide information on competitive dynamics. This yields a sample of auctions where a meaningful competition took place, which is essential for implementing our regression discontinuity approach.

5.2. Variable definitions

We construct key variables to capture firm outcomes and auction characteristics. First, we define a market such that two firms belong to the same market if they participated at least once in the same auction for the same type of good. This market classification is transitive, resulting in 7,581 distinct markets defined within at least one of our defined time periods.

Following Kawai et al. (2022), we construct an *Incumbent* dummy variable based on this market definition. Specifically, this variable equals 1 if a given firm won the last auction it participated in within the same market before the current auction, and 0 otherwise. This indicator reflects whether the firm entered the current tender as a reigning winner (incumbent) in the specific market or not.

Additionally, we define the *Last Bid* dummy variable equal to 1 if the firm was the last to submit a bid in the auction, and 0 otherwise.

Finally, we define the *Margin of Victory*, which serves as the running variable in our RDD design. For each auction, the margin of victory is calculated as the percentage difference between the winning bid and the second-lowest bid, expressed as a fraction of the winning bid. Formally, if b_{win} is the winning bid and $b_{\text{runner-up}}$ is the next lowest bid, the margin is $(b_{\text{runner-up}} - b_{\text{win}})/b_{\text{win}}$. When this margin is very small, the auction outcome was decided by a tiny price difference.

[1] An illustrative item is code 110639, which corresponds to "Furosemide 40mg tablets," classified under class 6531 (a category of pharmaceuticals) within group 65.

6. Empirical Results

Our empirical analysis identifies statistically significant discontinuities at the threshold of closely contested auctions, suggesting potential anti-competitive dynamics in public procurement markets.

Graphical representations of these effects are illustrated in Figures 1 and 2. Quantitative estimates supporting the graphical evidence are presented in Table 1. Figure 1 shows the estimated probability that the winning firm was the incumbent—defined as the firm winning the immediately preceding auction for a similar good—as a function of the bid margin. A clear discontinuity at the threshold (bid margin equal to zero) indicates that firms narrowly winning auctions (negative bid margin) exhibit a higher probability of incumbency compared to narrowly losing firms (positive bid margin). This pattern is consistent with incumbent firms benefiting disproportionately, which may be reflective of strategic bidding practices or bid rotation schemes.

Similarly, Figure 2 examines the probability that the winning firm submitted the last bid in the auction as a function of the bid margin. A comparable discontinuity at the cutoff suggests firms narrowly winning auctions have a higher likelihood of being the final bidder, potentially indicating strategic bid timing indicative of anti-competitive behavior, such as bid leakage or collusion.

The regression discontinuity estimates corresponding to these graphical results, accounting also for year and market fixed effects, are reported in Table 1. We find that being the incumbent is associated with a 1.82% increase in the probability of winning an auction within the same market. Similarly, winning firms are 6.36% more likely to submit the final bid. Both coefficients are statistically significant at the 1% level. The results remain robust to the inclusion of market and year fixed effects and statistically support the graphical evidence, reinforcing the interpretation of non-competitive forces at play in closely contested procurement auctions.

Figure 1: Probability of Incumbency around the Bid Margin

Presentations/rd_plot_won_t_minus_1_overall_market.pdf

Notes: The figure plots binned averages of incumbency status against the bid margin, where negative values correspond to narrowly won auctions and positive values reflect narrowly lost auctions. A distinct discontinuity at the cutoff suggests potential incumbency advantages.

Table 1: Effect of Winning Auction on Key Outcomes

	Incumbent		Last Bid	
	(1)	(2)	(3)	(4)
Winning Auction	0.0182***	0.0179***	0.0636***	0.0636***
	(0.005)	(0.005)	(0.008)	(0.008)
Observations	261370	260985	156418	156418
Optimal Bandwidth	0.02	0.02	0.01	0.01
Year FE		✓		✓
Market FE		✓		✓

Notes: Standard errors are reported in parentheses. *** indicates statistical significance at the 1% level. The dependent variables are defined as follows: (1) *Incumbent* is a dummy indicating whether the firm won the previous auction within the same market; (2) *Last Bid* is a dummy variable indicating whether the winning firm submitted the final bid in the auction.

Figure 2: Probability of Being the Last Bidder around the Bid Margin

Presentations/rd_plot_last_bid_overall_market.pdf

Notes: The figure plots binned averages of the probability that the winning firm submitted the last bid against the bid margin, where negative values correspond to narrowly won auctions and positive values reflect narrowly lost auctions. A discontinuity at the threshold indicates potential strategic timing in bid submissions.

7. Additional Results - TBC

7.1. Heterogenety by market

7.2. Correlation with predicted corruption

7.3. Firms outcomes

8. Conclusion

This paper tests a novel approach to detecting anti-competitive behavior in public procurement, applying it to auction data from São Paulo, Brazil. Inspired by Kawai et al. (2023), we use a regression discontinuity design around narrowly decided sealed-bid auctions to identify significant incumbency advantages and strategic bidding behavior. Specifically, narrowly winning firms are more likely to have won prior auctions and to submit the final bid, indicating favoritism or collusion inconsistent with purely competitive bidding.

These findings highlight vulnerabilities in the Brazilian procurement systems, suggesting that both corruption and firm collusion may distort competition. Finally, our analysis provides a practical tool for procurement oversight bodies to flag potential anti-competitive practices and improve accountability in public contracting.

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A. Appendix